

ABSTRACT

A method has been developed that attaches a free form deformation (FFD) lattice around an arbitrary object. This procedure has been tested by creating lattices for two different cases: the CRM wing and the CRM wing-body. Aerodynamic deformations are then tested to verify the versatility of free form deformation as a shape optimization tool.

BACKGROUND

- In traditional FFD, first a parallelepiped lattice is placed around an object which must be deformed. A coordinate transformation is then used to map every point on the object to a local (s,t,u) coordinate in the parallelepiped region. As lattice points are moved, Bezier curves and Bernstein polynomials are then used to cause a smooth deformation of the space and to the object.

$$\bar{X} = \sum_{j=0}^n \sum_{i=0}^m \sum_{k=0}^l B_i^m(s) B_j^n(t) B_k^l(u) \vec{P}_{ijk}$$

$$B_i^m(s) = \binom{m}{i} (1-s)^{m-i} s^i$$

- One drawback of standard FFD is that the lattice structure must be a parallelepiped. Thus, objects that are not rectangular in shape are not deformed elegantly.
- The optimization code currently is able to create a parallelepiped lattice around simple rectangular wings. However, to create the lattice around a more complicated shape, the user must analytically specify the locations of the points.

OBJECTIVE

An algorithm is desired that automatically places FFD lattice points around arbitrarily shaped objects.

METHODOLOGY

The new method for placing FFD points consists of two steps:

- To place the lattice points about any shape, first an axis direction is specified along which cross sections of the object will be studied. Using data from these cross sections, planes of FFD points are placed orthogonal to the axis that has been selected.
- Once the points are placed, the local coordinates of the object are found by using the Newton-Raphson method.

RESULTS

- The FFD point placement algorithm was tested on two cases; the CRM wing, and the CRM wing-body.
- The following four deformation cases were used to test on the CRM wing:
 - The sweep angle was increased by 15°
 - The dihedral angle was increased by 10°
 - The wing twist was adjusted
 - A winglet was added

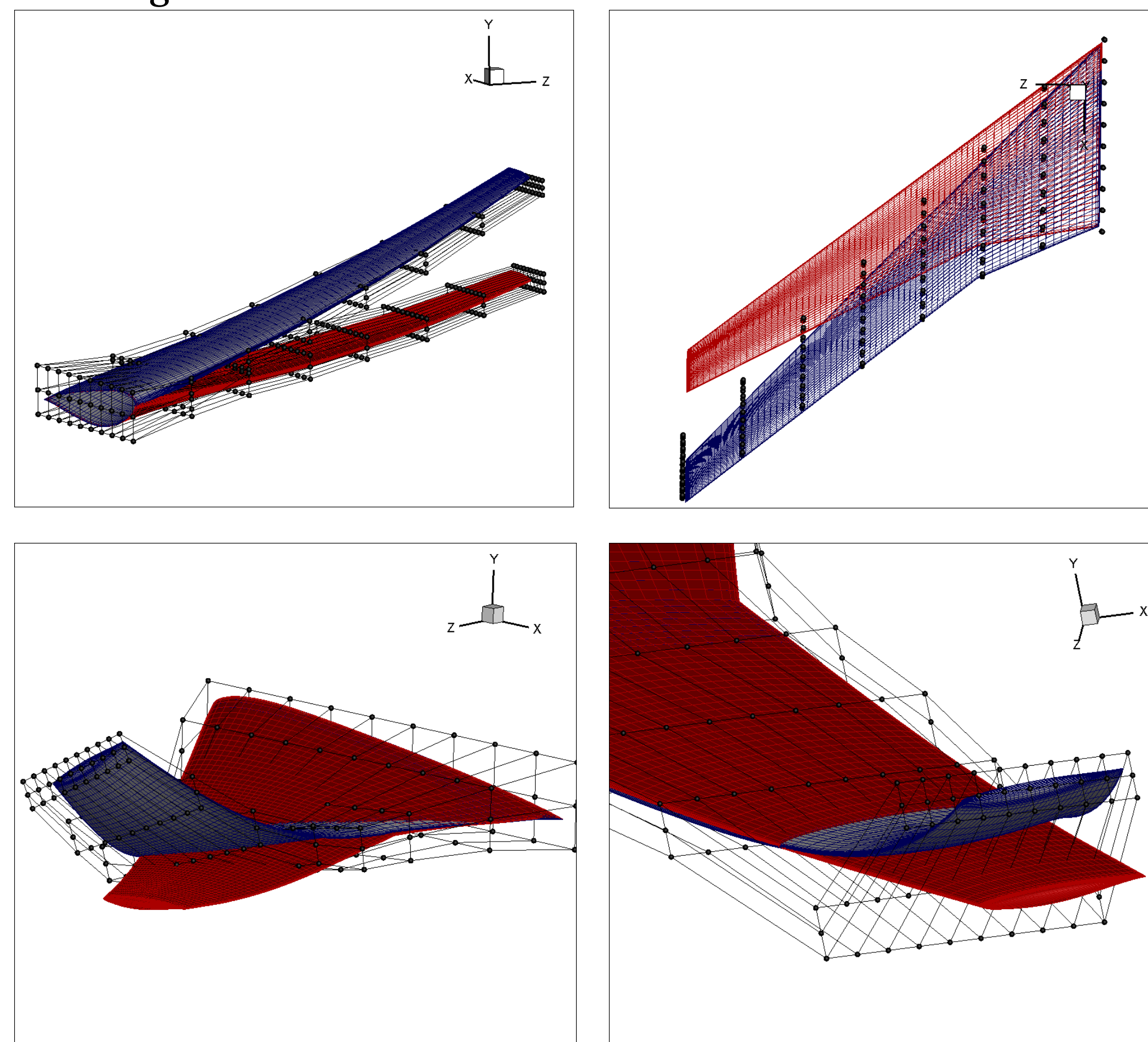
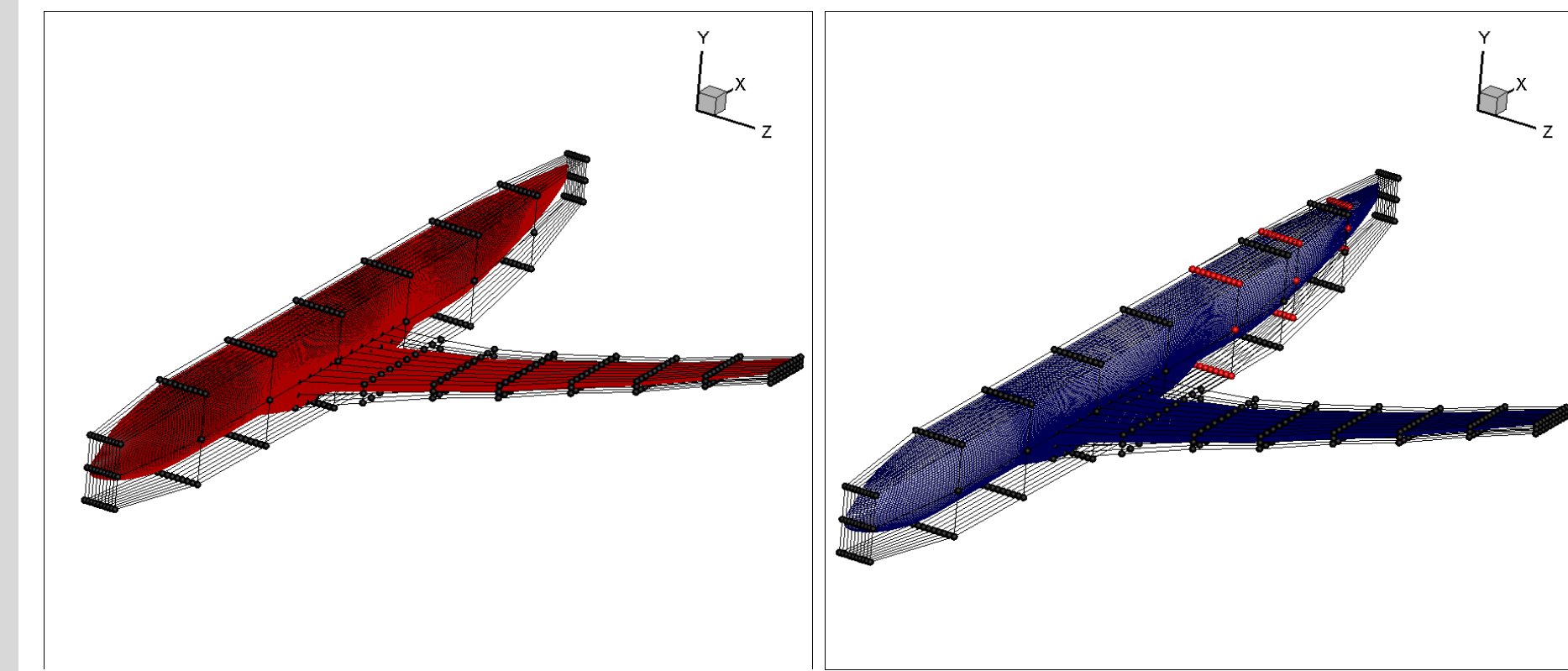


Figure 1: Results for the CRM Wing deformation. From top left and counter clockwise are results for (a) dihedral, (b) sweep, (c) winglet and (d) twist

RESULTS (Continued)

- The following deformation case was used to test the CRM wing-body:
 - The length of the fuselage was increased by 10%



CONCLUSIONS AND FURTHER WORK

- Both test cases show that the FFD lattice placement algorithm is able to create a lattice around the two different cases and smoothly deform them.
- A weakness of using Bezier curves to apply deformations is that it results in global deformations (a movement of any lattice point will result in a small deformation everywhere on the object), hence making it harder to impose geometric constraints. To solve this issue, the use of B-Splines and NURBS curves can be investigated.
- The new FFD algorithm's abilities must also be assessed by running it in the optimization code.

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REFERENCES

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